

Section 4H. Linear Discriminant Analysis (LDA)

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Recap

Because of the *Curse of Dimensionality*, we use parametric models to estimate the conditional class probabilities $\Pr[Y = k|X = \mathbf{x}]$ from a given dataset $\mathcal{D}$ (instead of local averaging)

Logistic regression is a particular parametric model in which we first pick a random input $x \sim f_X$ and assign a label to this input according to conditional class probabilities of the form:

$$p_1(x; \beta_0, \beta_1) = \Pr(Y = 1|X = x; \beta_0, \beta_1) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$
$$p_0(x; \beta_0, \beta_1) = \frac{1}{1 + e^{\beta_0 + \beta_1 x}}$$

However, this is not the only parametric model we can use...

Linear Discriminant Analysis

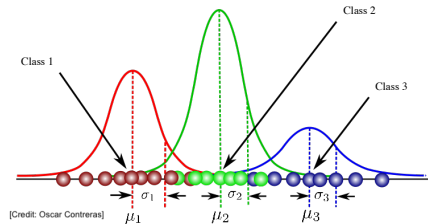
Linear Discriminant Analysis (LDA) assumes data is generated as follows:

1. For each sample point, we randomly draw an output label class k according to the following PMF (called the *prior* probabilities):

$$\Pr(Y = k) = \pi_k \text{ for } k = 1, \dots, K$$

2. Once an output class is chosen, we generate an input point $\mathbf{x}$ as follows

$$f_k(\mathbf{x}) = \Pr(X = \mathbf{x} | Y = k) = \mathcal{N}(\mu_k, \Sigma_k)$$



Linear Discriminant Analysis (cont.)

- ▶ Using Bayes theorem, we can compute the conditional class probabilities

$$\Pr(Y = k|X = \mathbf{x}) = \frac{\Pr(X = \mathbf{x}|Y = k) \Pr(Y = k)}{\sum_{\ell=1}^K \Pr(X = \mathbf{x}|Y = \ell) \Pr(Y = \ell)} = \frac{f_k(\mathbf{x}) \pi_k}{\sum_{\ell=1}^K f_{\ell}(\mathbf{x}) \pi_{\ell}}$$

- ▶ Hence, we can find the Bayes optimal classifier as follows:

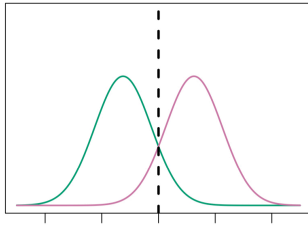
$$\begin{aligned} C(\mathbf{x}) &= \arg \max_k \Pr(Y = k|X = \mathbf{x}) \\ &= \arg \max_k \frac{f_k(\mathbf{x}) \pi_k}{\sum_{\ell=1}^K f_{\ell}(\mathbf{x}) \pi_{\ell}} = \arg \max_k f_k(\mathbf{x}) \pi_k \end{aligned}$$

Therefore, given an input $\mathbf{x}$, we classify it according to the class that gives the maximum value of $f_k(\mathbf{x}) \pi_k$

LDA Example

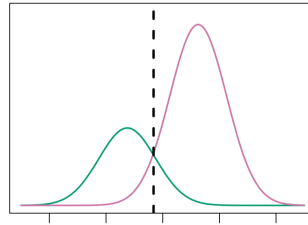
Example: Compare a binary classification problem with different priors

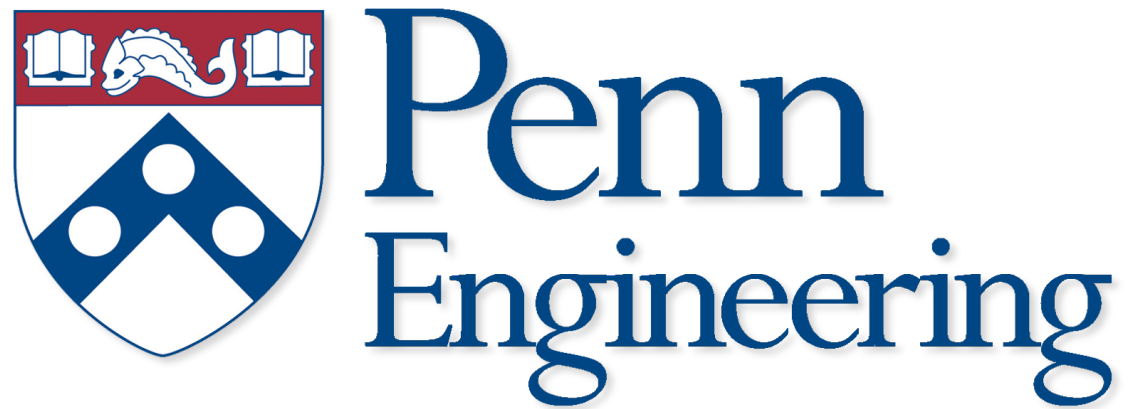
$$\pi_1=.5, \pi_2=.5$$



[Credit: Hastie and Tibshirani]

$$\pi_1=.3, \pi_2=.7$$





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